

Entry diagnostic — the Grade 9 toolkit

One lesson to find out exactly what each learner still owns from Grade 9, before anything new is built on it.

LESSON 1

1 × 40 MIN

ALGEBRA 10, REVISION

P1 · 1.1–1.8

LESSON CLOCK

3'

30'

5'

2'

Explain the purpose — this is not for a mark	3 min
The paper	30 min
Self-mark against the answers	5 min
Record the two weakest sections	2 min

LEARNING OBJECTIVES

- Factorise a quadratic and solve it.
- Use the formula and the discriminant.
- Simplify surds and index expressions.
- Read a learner’s gaps from the diagnostic and record them.

TEXTBOOK

National	Повторение, pp. 6–23
Cambridge	Pure Mathematics 1, pp. 1–23
Workbook	Diagnostic paper

01

Explanation

Why the year starts here

Grade 10 algebra spends its whole first quarter on **functions**. Almost every example is a quadratic, a linear expression or a surd. A learner who cannot factorise $x^2 - 5x + 6$ quickly will not be thinking about domain and range — they will still be thinking about factorising.

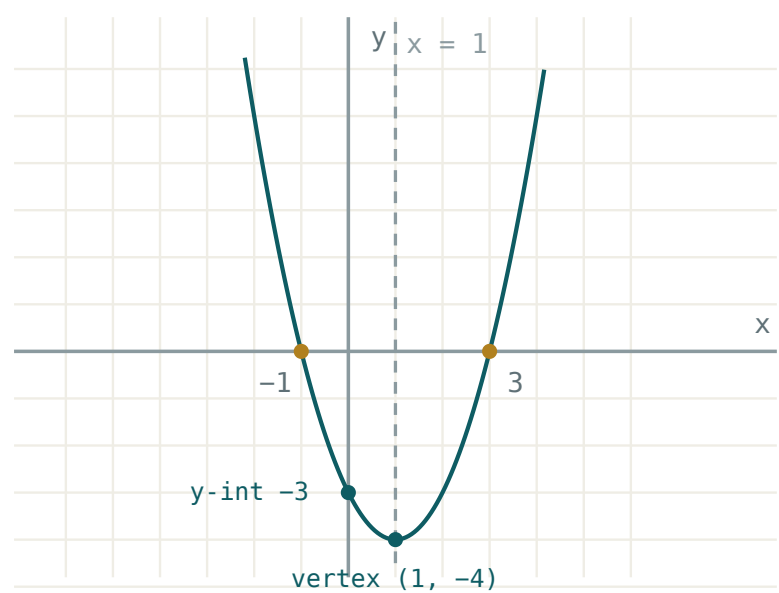
WHAT THIS LESSON IS FOR

Not a mark. A map. By the end of the lesson every learner should be able to name the **two** sections they were weakest in, and those two become their private revision list for the quarter.

Mark it together in the last five minutes, out of 20, in six sections of roughly equal weight.

The six sections

§	SKILL	MARKS	WHERE IT IS NEEDED THIS QUARTER
A	factorising a quadratic	4	roots, domain of a fraction
B	the quadratic formula and D	4	range, number of roots
C	surds and indices	3	domain of a root function
D	straight-line graphs	3	linear modelling
E	sketching a parabola	3	quadratic modelling
F	inequalities on a number line	3	writing a range



Everything in section E: the roots, the axis of symmetry, the vertex and the y-intercept of one parabola.

SAY THIS OUT LOUD

A low score today predicts nothing about the year. It only says which twenty minutes of revision will pay for themselves fastest.

02 Worked examples

EXAMPLE 1 Solve $x^2 - 5x + 6 = 0$ by factorising.

- ① Two numbers multiplying to 6 and adding to -5 .
They are -2 and -3 .

② $(x - 2)(x - 3) = 0$

③ $x = 2$ or $x = 3$

ANSWER

$x = 2, x = 3$

EXAMPLE 2 Find D for $2x^2 - 4x + 5 = 0$ and say how many real roots there are.

- ① $a = 2, b = -4, c = 5$
Read the signs carefully.

② $D = 16 - 4 \cdot 2 \cdot 5 = 16 - 40$

- ③ $D = -24 < 0$
No real roots.

ANSWER

$D = -24$, no real roots

EXAMPLE 3 Simplify $\sqrt{50} + \sqrt{18}$.

① $\sqrt{50} = 5\sqrt{2}$
 $50 = 25 \cdot 2.$

② $\sqrt{18} = 3\sqrt{2}$
 $18 = 9 \cdot 2.$

③ $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$
Like terms.

ANSWER

$$8\sqrt{2}$$

03 Interactive model

Work one question from each section on the board while the class self-marks.

Six questions, one from each section

$x^2 + 2x - 15$ factorises to:

For $x^2 - 6x + 9 = 0$, D is:

$\sqrt{27}$ in simplest form is:

The gradient of $2x + y = 6$ is:

2

-2

6

$\frac{1}{2}$

The vertex of $y = (x - 3)^2 + 1$ is:

$(3, 1)$

$(-3, 1)$

$(3, -1)$

$(1, 3)$

$-2x > 8$ gives:

$x > -4$

$x < -4$

$x > 4$

$x < 4$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Diagnostic test	Diagnostik test	Диагностический тест
Prior knowledge	Oldingi bilim	Ранее изученное
Factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Quadratic equation	Kvadrat tenglama	Квадратное уравнение
Discriminant	Diskriminant	Дискриминант
Surd	Irratsional ifoda	Иррациональное выражение
Index (exponent)	Daraja ko'rsatkichi	Показатель степени
Gap	Bo'shliq	Пробел
Baseline	Boshlang'ich daraja	Исходный уровень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$x^2 - 9$ factorises to:

If $D > 0$ a quadratic has:

$\sqrt{8} \cdot \sqrt{2}$ equals:

The line through $(0, 3)$ with gradient 2 is:

$$y = 2x + 3$$

$$y = 3x + 2$$

$$y = 2x - 3$$

$$y = 3x$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $x^2 - 7x + 12 = 0$

ANSWER $x = 3, x = 4$

2. Solve $x^2 - 16 = 0$

ANSWER $x = \pm 4$

3. Find D for $x^2 + 4x + 4 = 0$

ANSWER 0

4. Simplify $\sqrt{12}$

ANSWER $2\sqrt{3}$

5. Find the gradient of $y = 5 - 3x$

ANSWER -3

6. Where does $y = x^2 - 4$ cut the x -axis?

ANSWER $(\pm 2, 0)$

7. Solve $3x + 2 < 11$

ANSWER $x < 3$

Medium · the standard question

7 PROBLEMS

1. Solve $2x^2 - 5x - 3 = 0$

ANSWER $x = 3, x = -0.5$

2. Solve $x^2 + x - 12 = 0$

ANSWER $x = 3, x = -4$

3. Find D for $3x^2 - 2x + 1 = 0$ and say how many roots.

ANSWER $D = -8$, none

4. Simplify $\sqrt{75} - \sqrt{27}$

ANSWER $2\sqrt{3}$

5. Find the equation of the line through $(1, 5)$ and $(3, 11)$.

ANSWER $y = 3x + 2$

6. Write $y = x^2 - 6x + 5$ in completed-square form.

ANSWER $(x - 3)^2 - 4$

7. Solve $4 - 2x \geq 10$

ANSWER $x \leq -3$

Hard · stretch and reason

7 PROBLEMS

1. Solve $6x^2 + x - 2 = 0$

ANSWER $x = \frac{1}{2}, x = -\frac{2}{3}$

2. For what k does $x^2 + kx + 9 = 0$ have exactly one root?

ANSWER $k = \pm 6$

3. Simplify $\frac{6}{\sqrt{3}}$

ANSWER $2\sqrt{3}$

4. Find the vertex and roots of $y = x^2 - 4x + 3$.

ANSWER vertex $(2, -1)$, roots 1 and 3

5. Solve $x^2 - 5x + 6 > 0$

ANSWER $x < 2$ or $x > 3$

6. The line $y = 2x + k$ is a tangent to $y = x^2$. Find k .

ANSWER $x^2 - 2x - k = 0, D = 4 + 4k = 0$, so $k = -1$

7. Two numbers add to 12 and multiply to 35. Find them without trial.

ANSWER Roots of $t^2 - 12t + 35 = 0$: 5 and 7

07 Homework

Homework — 5 tasks

Only the two sections you were weakest in. Nothing else.

- Write down the two sections you scored lowest in.
- Five problems from the first of those two sections.
- Five problems from the second.
- Solve $x^2 - 3x - 10 = 0$ and check both roots by substitution.

5. Simplify $\sqrt{98} + \sqrt{50}$.